

## Foundation (Jalapeno)

- Q1.** A space station has 3 : 5 astronauts to engineers. If there are 9 astronauts, how many engineers are there?  
(A) 5  
(B) 6  
(C) 12  
(D) 15  
(E) 20
- Q2.** The ratio of oxygen to nitrogen in a planet's atmosphere is 1 : 4. If there are 5 parts of oxygen, how many parts of nitrogen are there?  
(A) 10  
(B) 15  
(C) 20  
(D) 25  
(E) 30
- Q3.** A scientist mixes 2 parts of chemical A with 3 parts of chemical B. If she uses 6 parts of chemical A, how many parts of chemical B will she use?  
(A) 6  
(B) 8  
(C) 9  
(D) 12  
(E) 15
- Q4.** The distance from Earth to Mars is 4 : 7 compared to the distance from Earth to the Sun. If the distance from Earth to the Sun is 93 million miles, what is the distance from Earth to Mars?  
(A) 40  
(B) 53  
(C) 53.14  
(D) 60
- (E) 80
- Q5.** A space probe travels 2 : 3 kilometers to meters. If it travels 6 kilometers, how many meters does it travel?  
(A) 6  
(B) 8  
(C) 9  
(D) 12  
(E) 16
- Q6.** The ratio of stars to planets in a galaxy is 5 : 8. If there are 20 stars, how many planets are there?  
(A) 24  
(B) 28  
(C) 32  
(D) 36  
(E) 40
- Q7.** A physicist mixes 3 parts of substance X with 2 parts of substance Y. If she uses 9 parts of substance X, how many parts of substance Y will she use?  
(A) 4  
(B) 5  
(C) 6  
(D) 8  
(E) 12
- Q8.** The ratio of time spent in space to time spent on Earth is 2 : 5 for an astronaut. If the astronaut spends 10 hours on Earth, how many hours does he spend in space?  
(A) 2  
(B) 3  
(C) 4  
(D) 6  
(E) 8

### Open Ended Questions (FRQ)

- Q9.** A rocket's fuel to oxygen ratio is 3 : 7. If 315 gallons of fuel are used, how many gallons of oxygen are needed? Show your work and explain your reasoning.
- Q10.** A planet's atmosphere has a ratio of 2 : 9 of carbon dioxide to nitrogen. If there are 18 parts of carbon dioxide, how many parts of nitrogen are there? Use cross-multiplication to solve the proportion and explain your answer.

### Chef's Tips

- Emphasize the importance of simplifying ratios
- Use visual aids to demonstrate proportions
- Encourage students to check their work using different methods